

The Gamma Distribution

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Introduction

The gamma distribution is a flexible right-skewed distribution on positive real numbers. It is parameterised by a shape and a rate (or scale) and includes the exponential, Erlang, and chi-squared as special cases. In applied work it models waiting times, concentrations, and response amplitudes – any positive continuous variable with light-to-moderate right skew.

Prerequisites

Exponential distribution, basic calculus.

Theory

A gamma random variable with shape $\alpha > 0$ and rate $\beta > 0$ has PDF

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0.$$

Scale parameterisation: $\theta = 1/\beta$, so $f(x) = \frac{1}{\theta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\theta}$.

Moments:

- $E[X] = \alpha/\beta$ (rate form) = $\alpha\theta$ (scale form).
- $\text{Var}(X) = \alpha/\beta^2 = \alpha\theta^2$.

Special cases:

- Shape $\alpha = 1$: exponential(β).
- Integer shape $\alpha = k$: Erlang distribution, which equals the sum of k iid exponentials with rate β .
- Shape $\alpha = \nu/2$, rate $\beta = 1/2$: chi-squared with ν df.

Sum property: independent $\text{Gamma}(\alpha_1, \beta)$ and $\text{Gamma}(\alpha_2, \beta)$ sum to $\text{Gamma}(\alpha_1 + \alpha_2, \beta)$.

Conjugacy: gamma is conjugate to the Poisson likelihood (prior on λ); gamma-Poisson mixture yields negative binomial.

Assumptions

Values must be positive. The flexibility comes from the two parameters; monotone hazard (increasing for $\alpha > 1$, decreasing for $\alpha < 1$, constant for $\alpha = 1$).

R Implementation

```
library(MASS)

shape <- 2.5; rate <- 0.8

X <- rgamma(1e5, shape = shape, rate = rate)
```

```

c(mean = mean(X), theoretical_mean = shape / rate,
  var = var(X), theoretical_var = shape / rate^2)

# Visualise the PDF
curve(dgamma(x, shape, rate), from = 0, to = 15,
  main = paste("Gamma(shape =", shape, ", rate =", rate, ")"),
  xlab = "x", ylab = "f(x)", col = "#2A9D8F", lwd = 2)

# MLE fit
fit <- fitdistr(X, "gamma")
fit$estimate

# Sum of independent exponentials equals Erlang (gamma with integer shape)
k <- 5; lambda <- 1
S <- replicate(5000, sum(rexp(k, rate = lambda)))
c(mean_empirical = mean(S), mean_theoretical = k / lambda,
  var_empirical = var(S), var_theoretical = k / lambda^2)

```

Output & Results

```

mean theoretical_mean      var theoretical_var
3.13                    3.125      3.92          3.906

      shape    rate
2.507    0.802      # MLE recovers true parameters

mean_empirical mean_theoretical var_empirical var_theoretical
      4.989           5.000          5.014          5.000

```

Empirical moments match theoretical; MLE recovers the true parameters; the sum of 5 iid exponentials has the predicted Erlang moments.

Interpretation

Gamma is a default parametric model for positive continuous variables that do not fit well to normal or log-normal. In generalised linear models, a gamma family (`glm(..., family = Gamma)`) is appropriate for positive continuous outcomes with variance proportional to the squared mean.

Practical Tips

- Watch R's `rgamma(n, shape, rate)` vs `rgamma(n, shape, scale = ...)`; only one of `rate` and `scale` should be given.
- Fitting a gamma to data: `MASS::fitdistr(x, "gamma")` or `fitdistrplus::fitdist`.
- Small shape ($\alpha < 1$) gives a distribution with a pole at zero; reasonable for heavily skewed data but breaks with zero values in a sample.
- For modelling variance in mixed models, the gamma distribution appears as the distribution of random-effect variances in certain hierarchical setups.
- Log-gamma (log of a gamma) is often approximately normal and easier to work with for some computations.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.

- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots